Nandana Dileep

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Experience

AI Engineer, JLR Technology & Business Services India July 2024 - Present

- Building **SARVAM**, a Git-native platform for requirements, design, and tests, adopted across **five vehicle programs** in India and the UK. Built CLI, VS Code, and web tools for authoring, diagram previews, and traceability.
- Built CI/CD gates for structural validation, AI semantic impact analysis, traceability checks, and reviewer approval; used a Bloom-filter ID registry to validate cross-repository references at merge time.
- Deployed **NVIDIA RAG Blueprint on AWS with seven NIM models** for requirements Q&A, impact analysis, and traceability insights, including Llama 3.1 8B and Nemotron document extraction.
- Built and maintain **OCTOPUS**, an ML test-prioritization system adopted across JLR; achieved **0.91 nDCG**, 0.88 ROC-AUC, and 0.86 normalized APFD, with recency-aware ranking to prevent test starvation.
- Built a **CatBoost rig-utilization classifier** on 500K+ daily telemetry events across approximately 200 rigs; achieved **94.4% F1** and **18% cost reduction**. Deployed inference through Kubernetes CronJobs.
- **JLR Global Techathon Winner, 2026:** built Meridian, an AI hiring platform for resume evidence mapping, skill-gap analysis, and adaptive technical assessments with human-review audit trails.

Software Development Engineer Intern, Amazon May 2023 - July 2023

- Designed and deployed an Alexa latency-analysis service using AWS Lambda, Coral, and OpenSearch.

Selected Projects

tiny-reranker - Inference Performance GitHub · Technical report · Medium article

- Ran 13 experiments on Apple M5 and NVIDIA L40S comparing PyTorch/MLX, compilation, custom Metal/CUDA/Triton kernels, batching, and Python/Rust HTTP coordination; separated operation, model-scoring, and request performance.
- Audited 11 implementations across 485 queries and validated 12,288 accepted Apple HTTP responses. Found that custom CUDA/Triton fusion did not beat compiled PyTorch on the matched L40S workload; retained reproducible protocols and negative results.

Mycelium - Agent Reliability Runtime GitHub · Documentation

- Created and maintain an open-source Python runtime for tool-input and authority checks, retries, concurrency, crash recovery, and uncertain outcomes; integrates with LangGraph, CrewAI, and custom agents.
- Implemented durable ledgers using file, SQLite, Redis, and PostgreSQL storage, sequential workflow replay, and an experimental HTTP sidecar for non-Python clients.

MLX Serving Engine on Apple Silicon GitHub · Architecture article

- Built an experimental MLX/FastAPI server with continuous batching, chunked prefill, prefix caching, educational paged KV-cache management, SSE streaming, and an OpenAI-compatible API; documented benchmark and implementation limits.

RefractorIQ - AI Code Analysis Platform GitHub

- Built Tree-sitter parsing, dependency graphs, duplicate detection, FAISS semantic search, and Gemini refactoring suggestions behind a FastAPI/Celery backend and React dashboard.

Identiti - Personal AI Memory GitHub | Live

- Built a React/Flask memory product extracting confidence-scored conversation entities into Neo4j, with Redis, Supabase authentication, a review interface, and portable export.

Technical Skills

Languages: Python, C++, TypeScript, SQL

AI / ML: PyTorch, Transformers, CUDA/Triton experiments, MLX, LangGraph, CrewAI, RAG, MCP

Backend / Data: FastAPI, Flask, PostgreSQL, Redis, Neo4j, OpenSearch, FAISS

Engineering: AWS, Azure, Docker, Kubernetes, Git, Linux, CI/CD, Pytest, React, Next.js

Education

Indian Institute of Technology Madras Aug 2020 - May 2024
Bachelor of Technology, Civil Engineering